



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

20-06-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : TEAS)

Test Type : REVIEW TEST

Test Pattern : JEE (Advanced)

TEST DATE : 20-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C,D	A,B,C,D	B,C	B,C	A,B	B,C,D	C
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	D	B	A	A	
SEC-3	Q.	1	2	3	4	5		
	A.	5	1	7	2	3		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C	A,B,D	A,B,D	B,D	A,B,D	B,C	A,B,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	D	B	B	C	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	6	3	9	8		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	A,B	B,C,D	B,D	B	A,B,C,D	A,B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	B	B	D	D	C	
SEC-3	Q.	1	2	3	4	5		
	A.	3	4	2	3	2		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (A,B,C,D)

$$B \xrightarrow{1-x} C \xrightarrow{x} B'$$

$$T = 0.5 \times 10 = 5 \text{ N}$$

$$\frac{3\lambda}{2} = 1 - x, \frac{5\lambda}{2} = x$$

$$\Rightarrow x = 62.5 \text{ cm}$$

$$\Rightarrow \lambda = \frac{2 \times 62.5}{5 \times 100} = 0.25 \text{ m}$$

$$\Rightarrow v = f \times \lambda = 500 \times 0.25 = 125 \text{ m/s}$$

$$\Rightarrow v = \sqrt{\frac{T}{\mu}} \Rightarrow \mu = 0.32 \text{ gm/m}$$

If we shift by 25 cm

x = 37.5 cm, 1 - x = 62.5 cm

Hence correct ans (A,B,C,D)

3.

Ans (B,C)

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + \frac{2\Delta r}{r}$$

$$V \cdot C \rightarrow L \cdot C = 1 \text{ MSD} - 1 \text{ VSD}$$

$$L \cdot C = 1 \text{ mm} - 0.9 \text{ mm} = 0.1 \text{ mm}$$

$$S \cdot G \rightarrow L \cdot C = \text{Pitch/no. of turn}$$

$$L \cdot C = 1 \text{ mm}/100 = 0.01 \text{ mm}$$

$$\text{for gold strip } \frac{\Delta R}{R} = \frac{5}{100}$$

In option (B)

$$\frac{\Delta R}{R} = \frac{0.1}{3+3 \times 0.1} + \frac{2 \times 0.1}{1+0}$$

$$= \frac{1}{33} + \frac{2}{100} = 0.08$$

In option (C)

$$\frac{\Delta R}{R} = \frac{0.1}{50+0} + \frac{2 \times 0.01}{0+0.01 \times 40}$$

$$= \frac{1}{500} + \frac{1}{20} = 0.052$$

4. **Ans (B,C)**

Target face = $10.4 \times 3.8 = 39.52$

ans will have 2 sig figure

Hence ≈ 40 is correct

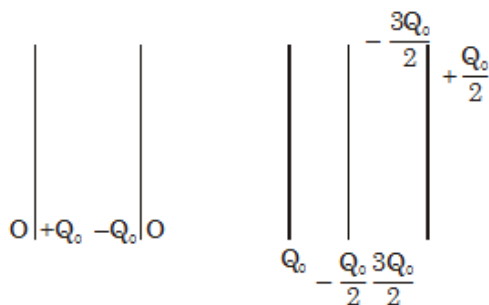
$\rightarrow$ Smallest face = $3.8 \times 1.02 = 3.876$

ans will have 2 sig figure

Hence $\approx 3.9 \text{ cm}^2$

$\rightarrow$ Volume $\approx 40 = 4.0 \times 10^1 \text{ cm}^3$

5. **Ans (A,B)**

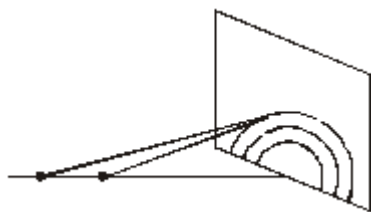


6. **Ans (B,C,D)**

In B,C sect up is like YDSE

Hence hyperbola

In (D)



PHYSICS

SEC-2

1. **Ans (B)**

$C = C_V + \frac{R}{1-x} \{PV^x = \text{constant}\}$

$\Delta\theta = nC\Delta T$

In B $x = 1.5 \Rightarrow C = \frac{3R}{2} - 2R$

$C = -ve$

2. **Ans (C)**

$C_{eq} = \frac{1 \times C_{V_m} + 1 \times C_{V_D}}{2}$

$= 2R$

$C = C_{eq} + \frac{R}{1-x}$

In (C) $\Rightarrow C = 2R - 2R = 0$

$\Rightarrow \Delta\theta = 0$

3. **Ans (D)**

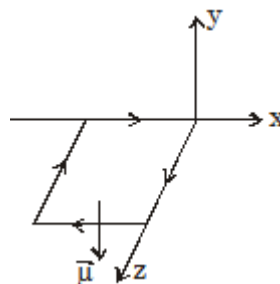
In (D)

$\uparrow P_f \rightarrow 2P_0$

$P/V = \text{constant} \Rightarrow P^2 = \text{constant } T$

as $P \uparrow \quad T \uparrow$

4. **Ans (B)**

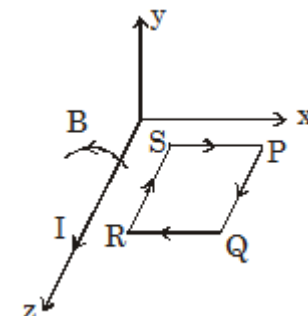


In (B)

$B \rightarrow$ in $-y$ direction

$\tau = \vec{\mu} \times \vec{B} = 0$

5. **Ans (A)**

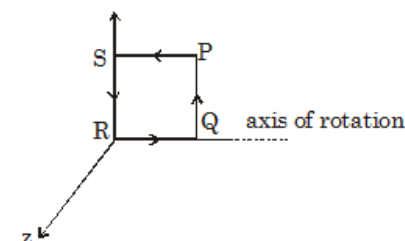


In option (A)

$F_{\text{wire-SR}} > F_{\text{wire-PQ}}$

$\vec{F}_{\text{net}} \neq 0$

6. **Ans (A)**



In option (A)

$B \rightarrow -y$ direction

$\tau_{PQ} = (PS) \times F_{SR} \neq 0$

PHYSICS

SEC-3

1. **Ans (5)**

$$f = f_0 \frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2 \dots (i)$$

$$E = E_0 \frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4 \dots (ii)$$

Solving (i) and (ii) $n = 2, z = 4$

$$L = mvr = \frac{nh}{2\pi}$$

$$\Delta L = \tau \Delta t = \Delta n \frac{h}{2\pi}$$

$$\tau = \frac{\Delta n}{\Delta t} \cdot \frac{h}{2\pi}$$

$$\tau \times 10^{27} = 15$$

2. **Ans (1)**

A B

N N

After 2 half life of B

$$\frac{N}{2^n} = \left(\frac{N}{2^2}\right) \frac{1}{2}$$

$$n = 3$$

3 half life of A has passed in 3 day $t_{1/2} A = 1$ day

3. **Ans (7)**

Loss in K.E of proton = Gain in potential energy of the proton - nucleus system

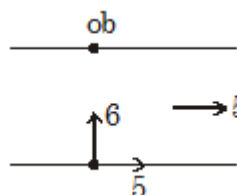
$$\frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\therefore \frac{p^2}{2m} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\therefore \frac{1}{2m} \left(\frac{h^2}{\lambda^2}\right) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\therefore \lambda = \sqrt{\frac{4\pi\epsilon_0 r \cdot h^2}{q_1 q_2 (2m)}} = 7\text{fm}$$

4. **Ans (2)**



$$f' = f_0 \left(\frac{336}{336 - 6} \right) = 112$$

$$\text{Beat} = |f' - f_0| = 2$$

5. **Ans (3)**

$$(2.31 - 0.31) \times 1632$$

3 sig digit

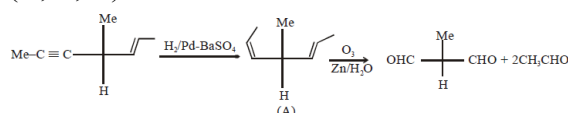
Hence ans will have 3 sig digit

CHEMISTRY

SEC-1

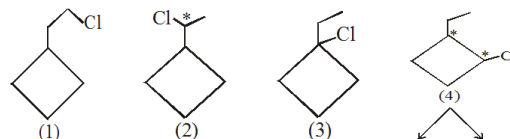
2. **Ans (A,B,D)**

(A, B, D)

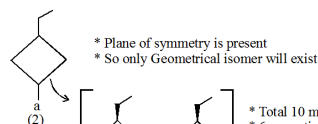
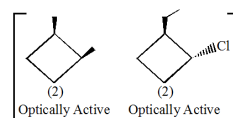


Compound A is optically active. Compound A when treated with $H_2(\text{excess})/Pt$, produces a compound which is optically inactive. Organic products formed in last step can reduce tollens reagent.

3. **Ans (A,B,D)**



(Chiral compound)
2 Stereo isomers



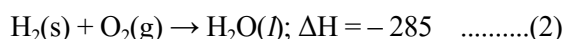
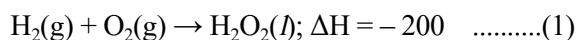
* Plane of symmetry is present
* So only Geometrical isomer will exist

ABD are correct option

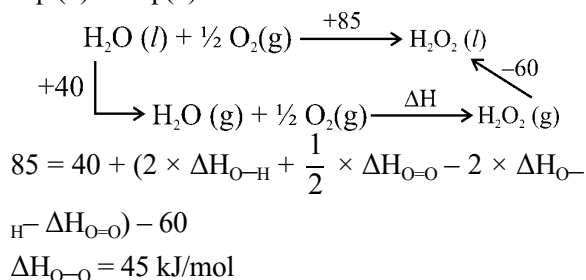
CHEMISTRY

SEC-3

4. **Ans (9)**



Eq. (1) & Eq (2)



MATHEMATICS

SEC-1

1. Ans (B,D)

Let $k = f(x) + \frac{1}{x}$. Then $k > 0$ (given)...(i)

so k will satisfy $f(x) \left[f\left(f(x) + \frac{1}{x}\right) \right] = 1$

$$\Rightarrow f(k) \left[f\left(f(k) + \frac{1}{k}\right) \right] = 1 \dots (ii)$$

$$\text{Also } f(x) \cdot f(k) = 1 \dots (iii)$$

by comparing (ii) and (iii) we say

$$f(x) = f\left(f(k) + \frac{1}{k}\right)$$

$$f(x) = f\left(\frac{1}{f(x)} + \frac{1}{f(x) + \frac{1}{x}}\right) \text{ (by i and iii)}$$

as f is injective function

$$x = \frac{1}{f(x)} + \frac{x}{xf(x) + 1}$$

$$x^2(f(x))^2 - xf(x) - 1 = 0$$

$$f(x) = \frac{x \pm \sqrt{x^2 + 4x^2}}{2x^2} = \frac{1 \pm \sqrt{5}}{2x}$$

$$\text{only } f(x) = \frac{1 - \sqrt{5}}{2x} \text{ satisfy all three given}$$

conditions.

2. Ans (A,B)

$$f(x) = \int_0^x f(t) \cos t dt - \int_0^x \cos(-t) dt$$

$$f'(x) = f(x) \cos x - \cos x$$

$$f'(x) - f(x) \cos x = -\cos x, \text{ I.F.} = e^{-\sin x}$$

$$e^{-\sin x} \cdot f(x) = \int e^{-\sin x} \cdot (-\cos x) dx$$

$$e^{-\sin x} \cdot f(x) = e^{-\sin x} + c$$

$$f(x) = 1 + c e^{\sin x}, \text{ Now } f(0) = 0 \Rightarrow c = -1$$

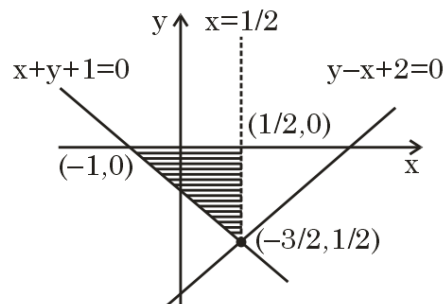
$$\Rightarrow f(x) = 1 - e^{\sin x}$$

3. Ans (B,C,D)

$$\text{Domain } x + y + 1 > 0,$$

$$y - x + 2 > 0$$

$$\text{Squaring } x + y + 1 < y - x + 2$$



$$\text{i.e. } x < \frac{1}{2}$$

Shaded region indicated in the figure

$$\text{and its area} = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{3}{2} = \frac{9}{8}$$

4. Ans (B,D)

$$I_m = \int_0^{m\pi} x |\sin x| e^{\cos 4x} dx \dots (1)$$

by using the property

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$I_m = \int_0^{m\pi} (m\pi - x) |\sin x| e^{\cos 4x} dx \dots (2)$$

adding eq. (1) & eq. (2) we get

$$2I_m = m\pi \int_0^{m\pi} |\sin x| e^{\cos 4x} dx$$

by using periodic property

$$2I_m = m^2 \pi \int_0^{\pi} |\sin x| e^{\cos 4x} dx$$

$$2I_m = 2m^2 \pi \int_0^{\pi/2} \sin x e^{\cos 4x} dx$$

$$\Rightarrow I_m = m^2 \pi \int_0^{\pi/2} \sin x e^{\cos 4x} dx \dots (3)$$

and

$$J_n = n \int_0^{\pi} |\cos x| e^{\cos 4x} dx = 2n \int_0^{\pi/2} \sin x e^{\cos 4x} dx \dots (4)$$

on dividing eq. (3) and (4)

$$\frac{I_m}{J_n} = \frac{m^2 \pi}{2n} \text{ now B,D is correct}$$

5. **Ans (B)**

We will rationalize the denominator,

$$I = \int_{-2014}^{2014} \frac{(1 + \sin^{2015}(x) - \sqrt{1 + \sin^{4030}(x)}) dx}{(1 + \sin^{2015}(x) + \sqrt{1 + \sin^{4030}(x)}) \times (1 + \sin^{2015}(x) - \sqrt{1 + \sin^{4030}(x)})}$$

$$= \int_{-2014}^{2014} \frac{(1 + \sin^{2015}(x) - \sqrt{1 + \sin^{4030}(x)})}{(1 + \sin^{2015}(x))^2 - (\sqrt{1 + \sin^{4030}(x)})^2} dx$$

$$= \int_{-2014}^{2014} \frac{(1 + \sin^{2015}(x) - \sqrt{1 + \sin^{4030}(x)})}{1 + (\sin^{2015}(x))^2 + 2\sin^{2015}(x) - 1 - \sin^{4030}(x)} dx$$

$$= \int_{-2014}^{2014} \frac{(1 + \sin^{2015}(x) - \sqrt{1 + \sin^{4030}(x)})}{2\sin^{2015}(x)} dx$$

$$I = \int_{-2014}^{2014} \frac{1}{2} (\sin x)^{-2015} dx + \int_{-2014}^{2014} \frac{1}{2} dx - \int_{-2014}^{2014} \frac{\sqrt{1 + \sin^{4030}(x)}}{\sin^{2015}(x)} dx$$

$$= I_1 + I_2$$

Using property if function is odd, $f(-x) = -f(x)$

$$\text{Then } \int_{-a}^a f(x) dx = 0$$

And $\sin(-x) = -\sin(x)$ {odd function}

Hence

$$I_1 = 0$$

$$\& I_2 = \int_{-2014}^{2014} \frac{\sqrt{1 + \sin^{4030}(x)}}{\sin^{2015}(x)} dx$$

Hence, $I_2 = 0$ {as $\sqrt{1 + \sin^{4030}(x)}$ is positive

& $\sin^{2015}(x)$ is odd}

$$\text{So, } I = \int_{-2014}^{2014} \frac{1}{2} dx$$

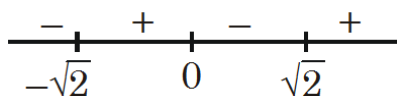
$$= \frac{1}{2} [x]_{-2014}^{2014}$$

$$I = \frac{(2014 + 2014)}{2} = 2014$$

Hence, option (B) is correct.

6. **Ans (A,B,C,D)**

$$f'(x) = \frac{8x(x - \sqrt{2})(x + \sqrt{2})}{(x^2 + 1)^3}$$



$f(x)_{\min}$ at $x = -\sqrt{2}, \sqrt{2}$

$$f(x)_{\min} = \frac{1}{3}$$

7. **Ans (A,B,C,D)**

$$\text{At } x = y = 0 : f(0) = 0$$

$$\text{At } x = 0; f(y) = f(-y) \Rightarrow f \text{ is even}$$

replace $y \rightarrow -y$:

$$f(x - y) = f(x + y) + f(x) \cdot f(-y)$$

$$\therefore f(x) \cdot f(y) = -f(x) \cdot f(-y)$$

$$\Rightarrow f(x) (f(y) + f(-y)) = 0$$

$$\Rightarrow f(x) = 0 \text{ or } f(y) = -f(-y) \text{ (f is odd)}$$

$\therefore$ either $f(x) = 0$ or $f(x)$ is simultaneously

odd and even.

$$\therefore f(x) = 0 \text{ only.}$$

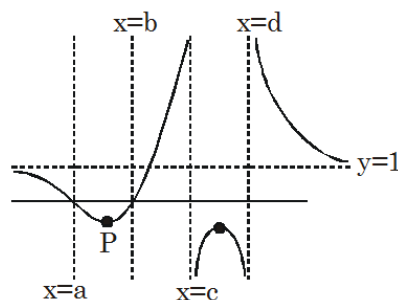
MATHEMATICS

SEC-2

1. **Ans (B)**

We cannot be sure about the coordinates of P

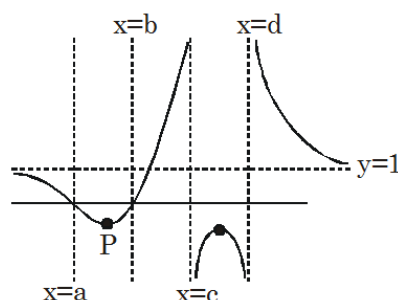
and Q as they depend on a, b, c and d



2. **Ans (B)**

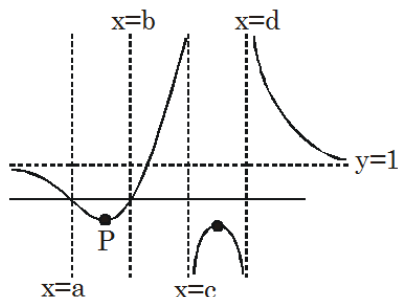
We cannot be sure about the coordinates of P

and Q as they depend on a, b, c and d



3. Ans (B)

We cannot be sure about the coordinates of P and Q as they depend on a, b, c and d



4. Ans (D)

$$\begin{aligned} \int_0^x \sqrt{1 - (f''(t))^2} dt &= \int_0^x f'(t) dt \Rightarrow \sqrt{1 - (f''(x))^2} \\ &= f'(x) \Rightarrow f''(x) = \pm \sqrt{1 - (f'(x))^2} \\ &\Rightarrow x = \pm \sin^{-1} f'(x) \Rightarrow f(x) = \pm (1 - \cos x) \\ &= \pm 2 \sin^2 \frac{x}{2} \\ f(x) &= 2 \sin^2 \frac{x}{2} \text{ or } f(x) = -2 \sin^2 \frac{x}{2} \text{ (reject)} \\ f(x) &\leq \frac{x^2}{2}, \forall x \in [0, 1] \end{aligned}$$

5. Ans (D)

$$\begin{aligned} \int_0^x \sqrt{1 - (f''(t))^2} dt &= \int_0^x f'(t) dt \Rightarrow \sqrt{1 - (f''(x))^2} \\ &= f'(x) \Rightarrow f''(x) = \pm \sqrt{1 - (f'(x))^2} \\ &\Rightarrow x = \pm \sin^{-1} f'(x) \Rightarrow f(x) = \pm (1 - \cos x) \\ &= \pm 2 \sin^2 \frac{x}{2} \\ f(x) &= 2 \sin^2 \frac{x}{2} \text{ or } f(x) = -2 \sin^2 \frac{x}{2} \text{ (reject)} \\ f(x) &\leq \frac{x^2}{2}, \forall x \in [0, 1] \end{aligned}$$

6. Ans (C)

$$\begin{aligned} \int_0^x \sqrt{1 - (f''(t))^2} dt &= \int_0^x f'(t) dt \Rightarrow \sqrt{1 - (f''(x))^2} \\ &= f'(x) \Rightarrow f''(x) = \pm \sqrt{1 - (f'(x))^2} \\ &\Rightarrow x = \pm \sin^{-1} f'(x) \Rightarrow f(x) = \pm (1 - \cos x) \\ &= \pm 2 \sin^2 \frac{x}{2} \\ f(x) &= 2 \sin^2 \frac{x}{2} \text{ or } f(x) = -2 \sin^2 \frac{x}{2} \text{ (reject)} \\ f(x) &\leq \frac{x^2}{2}, \forall x \in [0, 1] \end{aligned}$$

MATHEMATICS

SEC-3

1. Ans (3)

Replace x by $\frac{x-1}{x}$, we get

$$f\left(\frac{x-1}{x}\right) + f\left(\frac{1}{1-x}\right) = \tan^{-1}\left(\frac{x-1}{x}\right) \dots(2)$$

Replace x by $\frac{1}{1-x}$ in (1) we get

$$f\left(\frac{1}{1-x}\right) + f(x) = \tan^{-1}\left(\frac{1}{1-x}\right) \dots(3)$$

(1) - (2) + (3), we get

$$2f(x) = \tan^{-1}x + \tan^{-1}\left(\frac{1}{1-x}\right) - \tan^{-1}\left(\frac{x-1}{x}\right) \dots(4)$$

Also,

$$2f(1-x) = \tan^{-1}(1-x) + \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{x}{x-1}\right) \dots(5)$$

put $x = \frac{1}{2}$ and add (4) and (5)

$$\Rightarrow 4f\left(\frac{1}{2}\right) = \frac{3\pi}{2}$$

$$\Rightarrow N = f\left(\frac{1}{2}\right) = \frac{3\pi}{8}$$

2. Ans (4)

$$\begin{aligned} \left(\frac{x+2}{x+4}\right)^2 &= \frac{x^2+4x+4}{(x+4)^2} \\ &= \frac{x(x+4)}{(x+4)^2} + \frac{4}{(x+4)^2} \\ &= \underbrace{\frac{x}{x+4}}_{f(x)} + \underbrace{\frac{4}{(x+4)^2}}_{f'(x)} \end{aligned}$$

$$\text{as } f(x) = \frac{x}{x+4} \quad \text{so, } f'(x) = \frac{(x+4) - x}{(x+4)^2}$$

$$\therefore \int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

$$\begin{aligned} \text{So, } \int e^x \left(\frac{x+2}{x+4}\right)^2 dx &= \int e^x \left[\frac{x}{x+4} + \frac{4}{(x+4)^2}\right] dx \\ &= e^x \left(\frac{x}{x+4}\right) + C \end{aligned}$$

Hence, $\alpha = 0$ & $\beta = 4$

$$\text{Then } |\beta - \alpha| = |4 - 0| = 4$$

3. **Ans (2)**

$$\because \sqrt{x+y} + \sqrt{y-x} = 5 \dots(1)$$

$$\Rightarrow \frac{2x}{\sqrt{x+y} - \sqrt{y-x}} = 5$$

$$\sqrt{x+y} - \sqrt{y-x} = \frac{2x}{5} \dots\dots(2)$$

$$(1) + (2) \Rightarrow 2\sqrt{x+y} = 5 + \frac{2x}{5}$$

4. **Ans (3)**

$$= \lim_{y \rightarrow \infty} [(8y+3)(y+1)(y+2)$$

$$(y+3)(2y+3)(4y+5)(2y+7)]^{\frac{1}{7}} - 2y$$

$$= \lim_{y \rightarrow \infty} 2y \left[\left(1 + \frac{3}{8y}\right) \left(1 + \frac{1}{y}\right) \left(1 + \frac{2}{y}\right) \left(1 + \frac{3}{y}\right) \right.$$

$$\left. \left(1 + \frac{3}{2y}\right) \left(1 + \frac{5}{4y}\right) \left(1 + \frac{7}{2y}\right) \right]^{\frac{1}{7}} - 2y$$

$$= 2 \times \left(\frac{3}{8} + 1 + 2 + 3 + \frac{3}{2} + \frac{5}{4} + \frac{7}{2} \right) \times \frac{1}{7}$$

$$= 2 \times \frac{97}{56} = \frac{194}{56} \Rightarrow \left[\frac{A}{B} \right] = 3$$

5. **Ans (2)**

$$f(x) = \max \left(|x|, \left| \frac{x}{3} - 1 \right| \right)$$



2 points when $f(x)$ is non-derivable